

César Sauvage

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Education

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- Ph.D. at the National Center for Meteorological Research (CNRM)** Oct. 2016 – Dec. 2019
Paul Sabatier University, Toulouse, France
- Ocean-atmosphere coupling for prediction of Mediterranean heavy precipitation events : better modeling and impacts of river runoff and sea state.
- Engineering degree - Hydrology major** Sept. 2012 – Jan. 2015
Polytech Nice-Sophia Antipolis, France
- 3 year master program.
 - Semester in South Korea in 2014 at Incheon National University.

Positions Held

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- Oceanographic Researcher** Honolulu, HI, USA
University of Hawai'i at Mānoa Nov. 2024 – present
- Research Associate III** Woods Hole, MA, USA
Woods Hole Oceanographic Institution (WHOI) Jun. 2023 – Nov. 2024
- Postdoctoral Investigator** Woods Hole, MA, USA
Woods Hole Oceanographic Institution (WHOI) Feb. 2020 – Jun. 2023
- Advisor: Hyodae Seo and Carol Anne Calyson
- Ph.D. student** Toulouse, France
National Center for Meteorological Research (CNRM) Oct. 2016 – Dec. 2019
- Advisors: Cindy Lebeaupin Brossier, Marie-Noëlle Bouin and Véronique Ducrocq
- Guest student** Toulon, France
Mediterranean Institute of Oceanography (MIO) Apr. 2024 – Sept. 2015
- Advisors: Anne Molcard and Yann Ourmières

Publications

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- [9] - Sauvage, C., Seo, H., Zippel, S., Clayson, C-A., Edson, J., *Fetch-dependent Surface Wave Responses To Off-shore Wind Farms in the Northeast U.S. Coast*. Journal of Geophysical Research : Oceans, 130, e2024JC021162 [🔗](#), 2025.
- [8] - Seo, H., Sauvage, C., Renkl, C., Lundquist, J., Kirincich, A., *Sea Surface Warming and Ocean-to-Atmosphere Feedback Driven by Large-Scale Offshore Wind Farms under Seasonally Stratified Conditions*. Science Advances, Vol 11, Issue 45, [10.1126/sciadv.adw7603](#) [🔗](#), 2025.
- [7] - Nuijens, L., and coauthors including Sauvage, C., *The Air-Sea Interaction (ASI) submesoscale: physics and impact*. [10.5065/78ac-qd31](#) [🔗](#), 2024.
- [6] - Sauvage, C., Seo, H., Barr, B., Edson, J.B. and Clayson, C.A., *Misaligned Wind-Waves Behind Atmospheric Cold Fronts*. Journal of Geophysical Research : Oceans, 129, e2024JC021162 [🔗](#), 2024.
- [5] - Bouin M-N, C. Lebeaupin Brossier, S. Malardel, A. Voldoire, Sauvage, C., *The wave-age-dependent stress parameterisation (WASP) for momentum and heat turbulent fluxes at sea in SURFEX v8.1*, Geosci. Model Dev., 17, 117–141, [gmd-17-117-2024](#) [🔗](#), 2024.
- [4] - Sauvage, C., Seo, H., Clayson, C.A. and Edson, J.B., *Improving wave-based air-sea momentum flux parameterization in mixed seas*. Journal of Geophysical Research : Oceans, 128, e2022JC019277, [10.1029/2022JC019277](#) [🔗](#), 2023.

- [3] - **Sauvage, C.**, Lebeaupin Brossier, C., Bouin, M-N., *Towards kilometer-scale ocean-atmosphere-wave coupled forecast on a Mediterranean heavy precipitation event*. Atmos. Chem. Phys., 21, 11857–11887, [10.5194/acp-21-11857-2021](https://doi.org/10.5194/acp-21-11857-2021) [↗](#), 2021.
- [2] - **Sauvage, C.**, C.Lebeaupin Brossier, C., Bouin, M-N., and Ducrocq, V., *Characterization of the air-sea exchanges during a Mediterranean heavy precipitation event using realistic sea state modelling*. Atmos. Chem. Phys., 20, 1675–1699, [10.5194/acp-20-1675-2020](https://doi.org/10.5194/acp-20-1675-2020) [↗](#), 2020.
- [1] - **Sauvage, C.**, Lebeaupin Brossier, C., Ducrocq, V., Bouin, M-N., Vincendon, B., Verdecchia, M., Taupier-Letage, I. and Orain, F., *Impact of the representation of the freshwater river input in the Western Mediterranean Sea*. Ocean Modelling, 131, 115-131, [10.1016/j.ocemod.2018.09.005](https://doi.org/10.1016/j.ocemod.2018.09.005) [↗](#), 2018.

Honor and Awards

- Travel grant received to attend the **Workshop on Atmosphere-ocean coupling at (sub)mesoscales**, Leiden, the Netherlands, Sept. 2023.
- Travel grant received as early career scientist to attend the **US CLIVAR Mesoscale and Frontal-Scale Air-Sea Interactions Workshop**, Boulder, CO, USA, March 2023.
- Invited to participate to the **Physical Oceanography Dissertation Symposium XI (PODS)**, Kaua'i, Hawai'i. Oct. 2021

Teaching and Outreach

- **"Small Scale Air-Sea Interaction: Bulk Flux codes"**, Guest lecture in Oceanography class (OCN665) at University of Hawai'i at Mānoa.
- **"Ocean-Atmosphere Interactions Impact on Global Climate"**, Guest lecture in Earth and Science class at Glenville State University, Glenville, WV. Apr. 2023 (virtual).
- **"Introduction to ocean surface waves"**, Guest lecture for the MIT/WHOI Joint Program, Woods Hole, MA. Nov. 2021.

Student Mentoring

- **Summer 2025**, Co-advising Clint Reyes (University of Hawai'i at Mānoa). Clint is working on wave modeling of the typhoon Khanun passing near the islands of Okinawa during August 2023.
- **Winter 2024**, Co-advised Yoo-Jun Kim (Tokyo University) during November 2023-March 2024; Yoo-Jun is working on coastal and estuary circulation around Japan, looking into LES-scale winds and circulation.
- **Summer 2023**, Co-advised Peisen Tan (University of Miami); Peisen was working on laboratory and numerical modeling of short wind waves and swell interactions.

Community Activities

- **Session chair** : **"Waves in the large-scale air-sea system"** at The 4th workshop on Waves and Wave-Coupled Processes, Uppsala, Sweden, February 2023.
- **Session chair** : **"Regional / Shelf Scale"** at the Physical Oceanography Dissertation Symposium XI, Kaua'i, Hawai'i, USA, Oct. 2021.
- **Panelist reviewer** for the NOAA Modeling, Analysis, Predictions, and Projections (MAPP) Program.
- **Ad hoc reviewer** for the National Science Foundation (NSF) and for the French National Research Agency.
- **Reviewer** for Journal of Geophysical Research (JGR): Oceans; JGR: Atmospheres; Atmospheric Chemistry and Physics (ACP); Geoscientific Model Development (GMD); Scientific Reports; Limnology; Asia-Pacific Journal of Atmospheric Sciences (APAS); Journal of Physical Oceanography (JPO); Journal of Atmospheric and Oceanic Technology (JTECH); Geophysical Research Letters (GRL).
- **Co-organizer of the Physical Oceanography Department Seminar** at WHOI (2021- 2022).
- **Member of the Physical Oceanography Website Committee** at WHOI (2022 – 2024).